

Name:

Maths Class:

SYDNEY TECHNICAL HIGH SCHOOL



Year 12 Mathematics TRIAL HSC

August, 2015

Time allowed: 3 hours plus 5 minutes reading time

General Instructions:

- Reading time – 5 minutes
- Working time – 3 hours
- Marks for each question are indicated on the question.
- Approved calculators may be used
- All necessary working should be shown
- Full marks may not be awarded for careless work or illegible writing
- ***Begin each question on a new page***
- Write using black or blue pen
- All answers are to be in the writing booklet provided
- A set of Standard Integrals is provided at the rear of this Question Booklet, and may be removed at any time.

Total marks - 100

Section 1

10 Marks

- * Attempt Questions 1-10 on the sheet provided
- * Allow about 15 minutes for this section

Section II

90 marks

- Attempt Questions 11 – 16
- Allow about 2 hours and 45 minutes for this section

Section I

10 marks

Attempt Questions 1-10

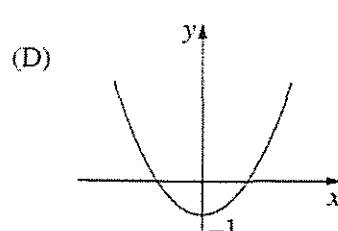
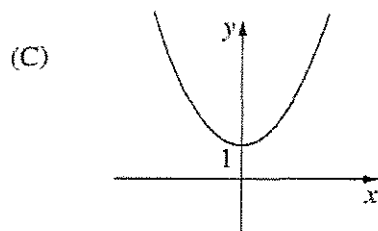
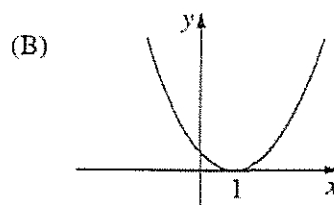
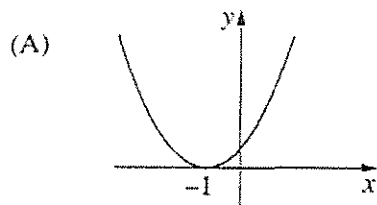
Allow about 15 minutes for this section.

Use the multiple-choice answer sheet for Questions 1-10

1. What is the value of $\frac{\sqrt[3]{3}}{2\pi}$, correct to 3 significant figures?

A. 0.23
B. 0.230
C. 0.229
D. 0.22

2. Which graph best represents $y = x^2 + 2x + 1$?



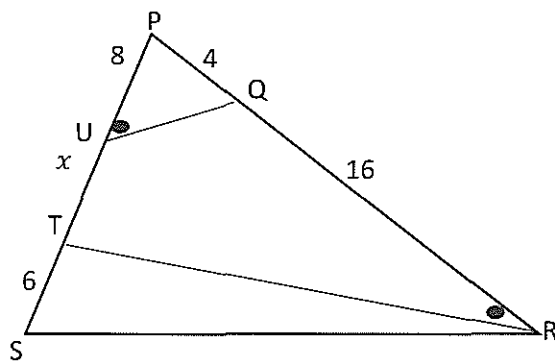
3. What is the solution to the equation $\log_3(x + 1) = 4$?

A. 11
B. 81
C. 80
D. 12

4. Which equation represents the line parallel to $2x - 3y = 8$, passing through the point $(-1, 2)$?

A. $3x + 2y - 1 = 0$
B. $3x + 2y - 8 = 0$
C. $2x - 3y - 8 = 0$
D. $2x - 3y + 8 = 0$

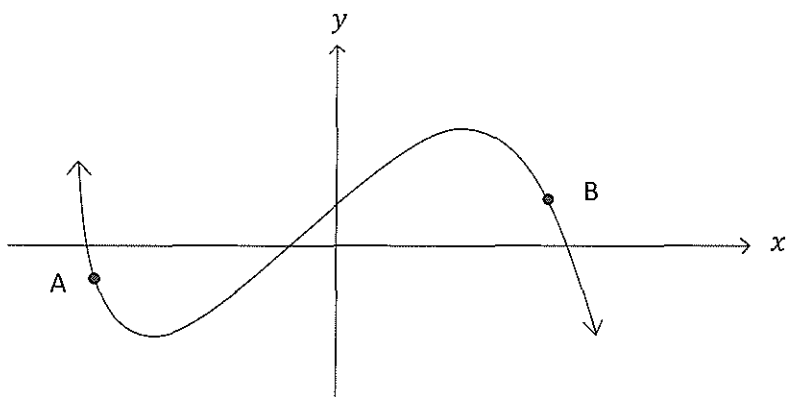
5. Which expression is a factorisation of $8x^3 - 27$?
- A. $(2x - 3)(4x^2 + 12x - 9)$
 B. $(2x + 3)(4x^2 - 12x + 9)$
 C. $(2x - 3)(4x^2 + 6x + 9)$
 D. $(2x + 3)(4x^2 - 6x + 9)$
6. The correct solutions to the equation $2\sin^2 x - 1 = 0$ for $-\pi \leq x \leq \pi$ are ?
- A. $\frac{\pi}{4}, \frac{3\pi}{4}$
 B. $\frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$
 C. $\pm \frac{\pi}{4}, \pm \frac{3\pi}{4}$
 D. $\pm \frac{\pi}{6}, \pm \frac{5\pi}{6}$
7. The value of $\sum_{n=0}^{\infty} 2 \times \left(\frac{3}{5}\right)^n$ is ?
- A. $\frac{6}{5}$
 B. 2
 C. 5
 D. 3
8. The length of PS in the following diagram is :



- A. 2
 B. 24
 C. 32
 D. 16

9. A parabola with a directrix $x = 2$ has a focus at $(-4, 3)$. The focal length of this parabola is?
- A. -6
 B. -3
 C. 6
 D. 3

10. For the curve $y = f(x)$, which of the following statements is correct?



- A. $f'(x) > 0$ at A and $f''(x) < 0$ at B
- B. $f'(x) < 0$ at A and $f''(x) < 0$ at B
- C. $f'(x) > 0$ at A and $f''(x) > 0$ at B
- D. $f'(x) < 0$ at A and $f''(x) > 0$ at B

End of Section 1

Section II

90 marks

Attempt Questions 11-16

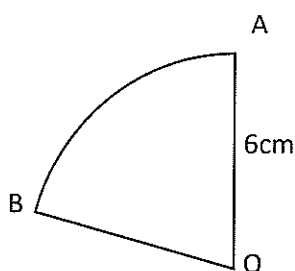
Allow about 2 hours and 45 minutes for this section.

Answer each question on the appropriate writing page. Extra pages are available.

In Questions 11-16, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks)

- a) Rationalise the denominator of $\frac{4}{\sqrt{6}-2}$ 2
- b) Factorise $9x^2 - 37x + 4$ 2
- c) Differentiate $\frac{x+1}{x^3}$ 2
- d) Find $\int \frac{dx}{(2x+1)^3}$ 2
- e) Find $\int \cos\left(\frac{x}{2}\right) dx$ 2
- f) Find the equation of the normal to the curve $y = 2\sqrt{x}$ at the point $x = 9$. 3
- g) The perimeter of the sector AOB is 15cm. Calculate the size of angle AOB, correct to the nearest degree. 2



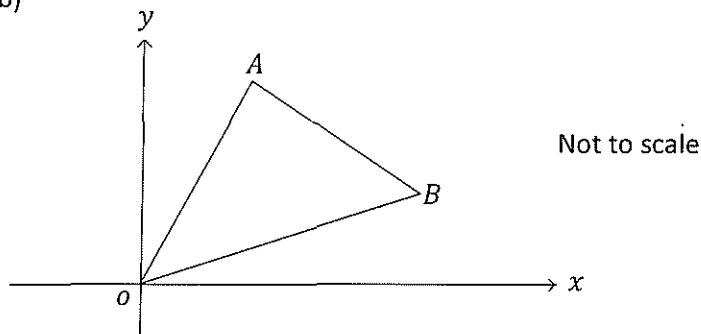
Question 12 (15 marks) Start a new page.

- a) Consider the series $106 + 97 + 88 + \dots$

Find the sum of all the positive terms belonging to this series.

3

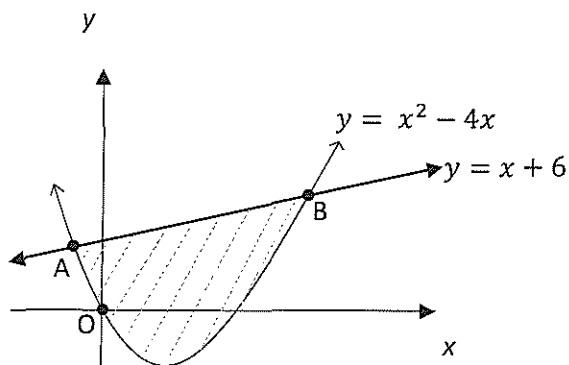
b)



Consider the points $A(3,5)$, $O(0,0)$ and $B(6,2)$ in the diagram above.

- I. Find the equation of BO. 2
- II. Show that the distance of the interval BO is $2\sqrt{10}$. 1
- III. Show that the area of triangle AOB is 12 square units. 2
- IV. Hence, or otherwise, find perpendicular distance from the point O to the line AB. 2

- c) The parabola $y = x^2 - 4x$ and the line $y = x + 6$ intersect at the points A and B.



- I. Find the x co-ordinate of the points A and B. 2
- II. Calculate the area enclosed by the parabola $y = x^2 - 4x$ and the line $y = x + 6$. 3

Question 13 (15 marks) *Start a new page.*

- a) i) Write down the exact value of $\tan 2x$ when $x = \frac{\pi}{6}$ 1
- ii) Give the exact value of $\int_0^{\frac{\pi}{6}} \sec^2 2x \, dx$ 2
- b) Simplify the expression $\frac{\cos(\frac{\pi}{2} - \theta)}{\sin(\pi + \theta)}$ 2
- c) Consider the curve $y = 2x^3 + 3x^2 - 12x - 9$
- i. Find the co-ordinates of any stationary points and determine their nature. 3
- ii. Show that a point of inflexion exists and state its co-ordinates. 2
- iii. Sketch the curve $y = f(x)$ in the domain $-3 \leq x \leq 3$, showing the
 y – intercept. 2
- iv. For what values of x , in the domain given in part (iii), is the curve both increasing
and concave down? 2
- v. Write down the minimum value for $y = f(x)$ in the interval $-3 \leq x \leq 3$. 1

Question 14 (15 marks) *Start a new page.*

a) i) Differentiate $2xe^{-x}$ 2

ii) Hence find $\int_0^1 xe^{-x} dx$ 2

b) The roots of the quadratic equation $3x^2 - kx + 18 = 0$ are α and β

i) Find the value of $\alpha\beta$. 1

ii) Given that $\alpha^2 + \beta^2 = 4$, find the value/s of k . 2

c) The region bounded by the curve $y = 1 + x^2$ and the x -axis between

$x = 0$ and $x = 3$ is rotated about the x -axis to form a solid. Find the

volume of the solid. 3

d) The quantity of bacteria in a culture is growing according to the equation

$$\frac{dM}{dt} = kM$$

where M is the mass of the bacteria present in the culture in mg, t is the time in

hours and k is a constant.

i) Show that $M = Ae^{kt}$ is a solution to the equation, where A is a constant. 1

ii) The time for the bacteria to double in mass is calculated to be 12 minutes.

Write down the value of k , correct to 4 significant figures. 2

iii) If the initial amount of bacteria is 8mg, after how many minutes does the

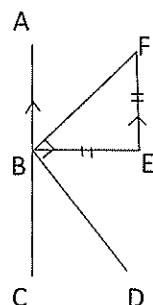
bacteria reach 1 gram? (Answer to one decimal place). 2

Question 15 (15 marks) *Start a new page.*

a) Solve $2 \log_5 x - \log_5(x + 2) = \frac{2}{3} \log_5 125$

3

b)



In the diagram BEF is a triangle with $BE = EF$. BF is perpendicular to BD and the line AC through B is parallel to EF.

i) Copy the diagram into your answer booklet.

ii) Prove that BF bisects angle ABE.

2

iii) Prove that BD bisects angle EBC.

2

c) Water is being released from a rainwater tank. The rate of flow, R litres per minute is given by $R = t(t - 12)^2$, where t is the number of minutes since the water began to flow.

i) For how long does the water flow?

1

ii) Find the maximum rate of flow.

2

iv) What is the total volume of water released from the tank?

3

d) Given $y = e^{kx}$, find the value of k such that $y = 2 \frac{dy}{dx} - \frac{d^2y}{dx^2}$

2

Question 16 (15 marks) *Start a new page.*

- a) Allied Lending is offering a special on loans of \$50,000.00 or more. The terms offered are a reducible interest rate of only $\frac{1}{2}\%$ per month with the first six months interest free.

Paddington takes out a loan of \$80,000.00 to start his marmalade shop and agrees to the terms set out by Allied Lending.

Paddington agrees to repay the loan in equal monthly instalments of \$M, over 10 years with the first repayment due at the end of the first month. Let A_n be the amount owing at the end of the n th repayment.

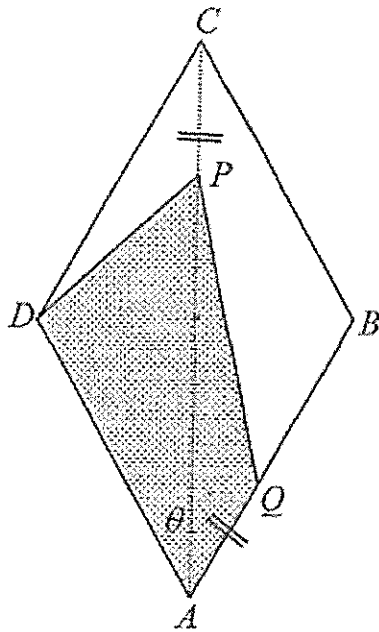
- i) Write down an expression for the amount Paddington owes at the end of the first six months. 1
- ii) If the interest is calculated immediately before each repayment is made, show that the amount owing at the end of 8 months is,
- $$A_8 = (80000 - 6M)(1.005)^2 - M(1.005 + 1) \quad 2$$
- iii) Hence show that $A_{120} = (80000 - 6M)(1.005)^{114} - 200M(1.005^{114} - 1)$ 2
- iv) Find the value of each monthly repayment correct to the nearest dollar. 2

b) ABCD is a rhombus of side 2cm.

P and Q are points on AC and AB respectively such that

$CP = AQ = x \text{ cm}$. $\angle DAP = \theta$ (where $0 < \theta < \frac{\pi}{2}$) and θ is a constant.

Let the area of the shaded area PDAQ be $S \text{ cm}^2$.



- | | | |
|-------|---|---|
| (i) | Show that $S = \frac{\sin \theta}{2} (4 \cos \theta - x)(2 + x)$ | 3 |
| (ii) | If $\frac{dS}{dx} = 0$, find x in terms of θ | 2 |
| (iii) | Find $\frac{d^2S}{dx^2}$ in terms of θ | 1 |
| (iv) | Suppose that $\theta = \frac{\pi}{6}$, show that S attains its maximum when $\frac{PC}{AC} = \frac{\sqrt{3}-1}{2\sqrt{3}}$ | 2 |

End of Assessment task

STHS 2unit Trial

Solutions

$$e) \int \cos\left(\frac{1}{2}x\right) dx \\ = 2\sin\left(\frac{x}{2}\right) + C$$

$$f) x = 9 \quad y = 2\sqrt{9} \quad \therefore (9, 6) \\ = 6$$

$$y' = x^{-1/2} \\ m_T = \frac{1}{\sqrt{9}} \\ = \frac{1}{3} \quad \therefore M_N = -3$$

$$\text{Eq: } y - 6 = -3(x - 9) \\ y - 6 = -3x + 27 \\ 3x + y - 33 = 0$$

Question 11

$$a) \frac{4}{\sqrt{6}-2} \times \frac{\sqrt{6}+2}{\sqrt{6}+2} \\ = \frac{4(\sqrt{6}+2)}{6-4}$$

$$= 2(\sqrt{6}+2) \quad \leftarrow \text{either} \\ = 2\sqrt{6} + 4 \quad \leftarrow$$

$$g) \rho = r\theta + 12$$

$$r\theta = 3$$

$$6\theta = 3$$

$$\theta = \frac{1}{2}$$

$$\therefore \theta = \frac{1}{2} \times \frac{180}{\pi}$$

$$= 29^\circ$$

$$b) 9x^2 - 37x + 4 \\ = (9x-1)(x-4)$$

$$c) \frac{x+1}{x^3} = x^{-2} + x^{-3}$$

$$\frac{d}{dx} = -2x^{-3} - 3x^{-4}$$

$$= -\frac{2}{x^3} - \frac{3}{x^4}$$

$$d) \int \frac{dx}{(2x+1)^3} = \int (2x+1)^{-3} dx$$

$$= \frac{(2x+1)^{-2}}{-2 \times 2} + C$$

$$= \frac{-1}{4(2x+1)^2} + C$$

Question 12

a) $106 + 97 + 88 + \dots$

AP $a = 106$

$d = -9$

$T_n < 0$

$106 + (n-1)(-9) < 0$

$-9n < -715$

$n > 12.7$

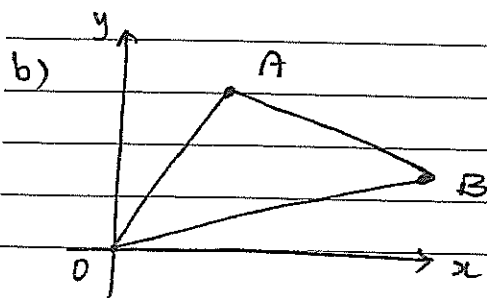
$\therefore n = 13$

$T_{12} = 7$

$S_n = \frac{n}{2}(a+l) \text{ or } \frac{n}{2}(2a+(n-1)d)$

$= \frac{12}{2}(106+7)$

$= 678$



(i) Equation $M_{BO} = \frac{0-2}{0-6}$

$= \frac{1}{3}$

$\therefore y = \frac{1}{3}x \rightarrow 3y = x$

(ii) $B(6,2)$ $O(0,0)$

$d_{BO} = \sqrt{36+4}$

$= \sqrt{40}$

$= 2\sqrt{10}$

(iii) $A(3,5)$ $d_1 = 12/\sqrt{10}$

$A = \frac{1}{2}hb$

$= \frac{1}{2} \times 12/\sqrt{10} \times 2\sqrt{10}$

$= 12 u^2$

AB $(3,5)$ $(6,2)$

$d_{AB} = \sqrt{(6-3)^2 + (2-5)^2}$

$= \sqrt{9+9}$

$= \sqrt{18}$

$= 3\sqrt{2}$

$\therefore \frac{1}{2}hb = \frac{1}{2} \times h \times 3\sqrt{2}$

$12 = \frac{1}{2} \times h \times 3\sqrt{2}$

$8 = h\sqrt{2}$

$h = \frac{8}{\sqrt{2}} \quad (4\sqrt{2})$

c) Solving

$x^2 - 4x = x + 6$

$x^2 - 5x - 6 = 0$

$(x-6)(x+1) = 0$

$x = 6, x = -1$

(i) $A = \int_{-1}^6 (x+6) - (x^2-4x) dx$

$= \int_{-1}^6 (5x+6-x^2) dx$

$= \left[\frac{5x^2}{2} + 6x - \frac{x^3}{3} \right]_{-1}^6$

$= \frac{5}{2}(36) + 36 - \frac{6^3}{3} - \left(\frac{5}{2} - 6 + \frac{1}{3} \right)$

$= 57 \frac{1}{6} u^2$

Question 13

$$\begin{aligned} \text{i) } \tan 2x &= \tan 2(\pi/6) \\ &= \tan \pi/3 \\ &= \sqrt{3} \end{aligned}$$

$$\begin{aligned} \text{ii. inflexion } y'' &= 0 \\ 12x + 6 &= 0 \\ x &= -1/2 \\ y &= -5/2 \end{aligned}$$

$$\begin{aligned} \text{ii) } \int_0^{\pi/6} \sec^2 2x \, dx &= \left[\frac{1}{2} \tan 2x \right]_0^{\pi/6} \\ &= \frac{1}{2} [\tan \pi/3 - \tan 0] \\ &= \frac{1}{2} [\sqrt{3} - 0] \\ &= \sqrt{3}/2 \end{aligned}$$

$$\begin{aligned} \text{concavity check} \\ x &= -1 \quad x = -1/2 \quad x = 0 \\ y'' &= -6 \quad 0 \quad y'' = 6 \end{aligned}$$

$\therefore$ change in concavity @ $(-1/2, -5/2)$

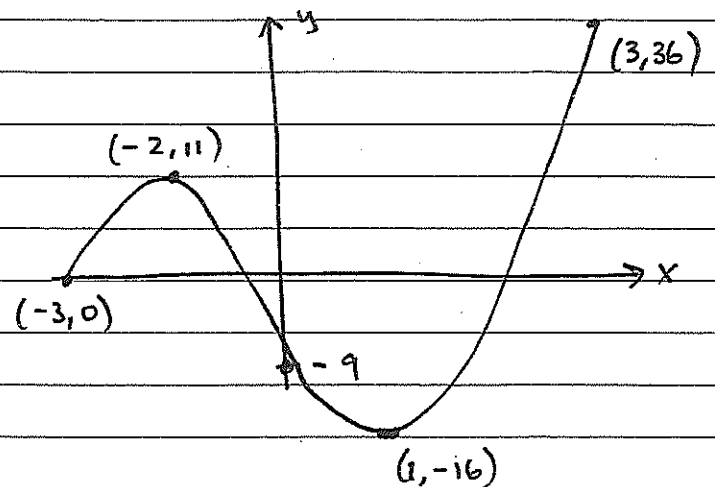
iii. end pts $(-3, 0)$ & $(3, 36)$

$$\begin{aligned} \text{b) } \frac{\cos(\pi/2 - \theta)}{\sin(\pi + \theta)} &= \frac{\sin \theta}{-\sin \theta} \\ &= -1 \end{aligned}$$

$$\text{c) } y = 2x^3 + 3x^2 - 12x - 9$$

$$dy/dx = 6x^2 + 6x - 12$$

$$d^2y/dx^2 = 12x + 6$$



i) Stat pts

$$6(x^2 + x - 2) = 0$$

$$6(x+2)(x-1) = 0$$

$$\begin{array}{l|l} x = -2 & x = 1 \\ y = 11 & y = -16 \end{array}$$

$$\therefore (-2, 11) \text{ \& } (1, -16)$$

Nature

x	-3	-2	-1
y'	15	0	-12

x	-1	1	2
y'	-12	0	24

$$\therefore \text{MAX TP } (-2, 11) \quad \& \quad \text{MIN TP } (1, -16)$$

iv) $\nearrow$ and $\searrow$ $-3 \leq x < -2$

accept $-3 < x < -2$

do not accept $-3 \leq x \leq -2$

v) min value is -16

do not accept $(1, -16)$

Question 14

a) $\frac{d}{dx}(2xe^{-x})$

$$= 2e^{-x} - 2xe^{-x}$$

$$= \pi \left[x + \frac{2x^3}{3} + \frac{x^5}{5} \right]_0^3$$

$$= \pi \left[3 + \frac{2}{3} \cdot 27 + \frac{3^5}{5} - (0) \right]$$

ii) $\frac{1}{2} \frac{d}{dx}(2xe^{-x}) = e^{-x} - xe^{-x}$

$$\therefore \int xe^{-x} dx = \left[\int e^{-x} - \int \frac{1}{2} \frac{d}{dx}(2xe^{-x}) \right] dx = \frac{348\pi}{5} u^3$$

$$= \int e^{-x} dx - xe^{-x}$$

$$= -e^{-x} - xe^{-x}$$

d) $m = Ae^{kt}$

$$\frac{dm}{dt} = k \cdot Ae^{kt}$$

$$= k \cdot m \quad \therefore \text{a sol}^n$$

$$\therefore \int_0^1 xe^{-x} dx = \left[-e^{-x} - xe^{-x} \right]_0^1$$

$$= -e^{-1} - 1e^{-1} - (-e^0 - 0)$$

$$= -1/e - 1/e + 1$$

$$= 1 - 2/e$$

$$= \boxed{1 - \frac{2}{e}}$$

ii. $2A = Ae^{k(12)}$

$$2 = e^{\frac{1}{2}k}$$

$$2 = e^{12k}$$

$$\ln 2 = \frac{1}{2}k$$

$$\ln 2 = 12k$$

$$k = 3.466$$

$$k = \frac{1}{12} \ln 2$$

$$\approx 0.05776 \quad (4 \text{ sig fig})$$

b) i. $\alpha\beta = c/a$

$$= 6$$

iii. $1 \text{ gram} = 1000 \text{ mg}$

$$1000 = 8e^{kt}$$

$$\text{using } k = 0.05776$$

$$125 = e^{kt}$$

$$\ln 125 = kt$$

$$t = \ln 125 \div k$$

$$= 83.592 \dots$$

ii. $\alpha^2 + \beta^2 = 4$

$$\alpha + \beta = \frac{k}{3}$$

$$(\alpha + \beta)^2 - 2\alpha\beta = 4$$

$$(\alpha + \beta)^2 - 12 = 4$$

$$\left(\frac{k}{3}\right)^2 = 16$$

$$\frac{k}{3} = 4 \text{ or } -4$$

$$\therefore k = \pm 12$$

$$\therefore 83.6 \text{ minutes.}$$

c) $V_x = \pi \int y^2 dx$

$$= \pi \int_0^3 (1 + x^2)^2 dx$$

$$= \pi \int_0^3 1 + 2x^2 + x^4 dx$$

Question 15

$$a) 2\log_5 x - \log_5(x+2) = \frac{2}{3}\log_5 125$$

$$\log_5 \left(\frac{x^2}{x+2} \right) = \log_5 (5^3)^{2/3}$$

$$\log_5 \left(\frac{x^2}{x+2} \right) = \log_5 5^2$$

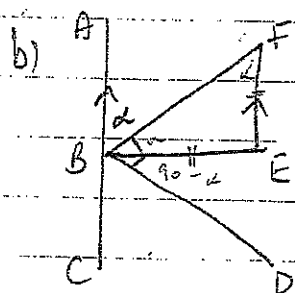
$$\frac{x^2}{x+2} = 25$$

$$x^2 - 25x - 50 = 0$$

$$x = \frac{25 \pm \sqrt{825}}{2}$$

$$\text{as } x > 0$$

$$x = \frac{25 + \sqrt{825}}{2} \text{ only}$$



$$1. \text{ let } \angle ABF = \alpha$$

$$\angle BFE = \angle ABF \text{ (alternate angles } AC \parallel EF)$$

$$= \alpha$$

$$\angle FBE = \angle BFE \text{ (equal angles opposite equal sides } FE = BE \text{ given)}$$

$$= \alpha$$

$$\therefore \angle ABF = \angle FBE (= \alpha)$$

$$\therefore BF \text{ bisects angle ABE}$$

$$1. \angle EBD = 90 - \alpha \text{ (adjacent complementary with } \angle FBE)$$

$$\angle ABE + \angle EBD + \angle CBD = 180 \text{ (straight line)}$$

$$\therefore \angle CBD = 90 - \alpha$$

$$= \angle EBD$$

and BD bisects $\angle EBC$

$$c) R = \frac{dv}{dt} = t(t-12)^2$$

$$1) R > 0 \therefore 0 \leq t \leq 12$$

$\therefore$ flows for 12 minutes

$$11) \text{ MAX flow rate } \frac{dR}{dt} = 0$$

$$R = t(t^2 - 24t + 144)$$

$$= t^3 - 24t^2 + 144t$$

$$\frac{dR}{dt} = 3t^2 - 48t + 144 = 0$$

$$t^2 - 16t + 48 = 0$$

$$(t-12)(t-4) = 0$$

$$\text{as } 0 \leq t \leq 12$$

$$\text{test } t = 4$$

t	3	4	5
$\frac{dR}{dt}$	27	0	-21

$$\text{max flow rate is } R = 4(4-12)^2 = 256 \text{ L/min}$$

$$111) V = \int_0^{12} t^3 - 24t^2 + 144t \, dt$$

$$= \left[\frac{t^4}{4} - 8t^3 + 72t^2 \right]_0^{12}$$

$$= \left[\frac{12^4}{4} - 8(12)^3 + 72(12)^2 - 0 \right]$$

$$= 1728 \text{ L}$$

$$d) y = e^{kx} \quad y' = ke^{kx} \quad y'' = k^2 e^{kx}$$

$$y = 2y' - y''$$

$$e^{kx} = 2(ke^{kx}) - (k^2 e^{kx})$$

$$1 = 2k - k^2$$

$$k^2 - 2k + 1 = 0$$

$$(k-1)^2 = 0$$

$$\therefore k = 1$$

Question 16

a)

$$(i) A_6 = 80000 - 6M$$

$$(ii) A_7 = (80000 - 6M)(1.005) - M$$

$$A_8 = [(80000 - 6M)(1.005) - M](1.005) - M$$

$$= (80000 - 6M)(1.005)^2 - M(1.005) - M$$

$$= (80000 - 6M)(1.005)^2 - M[1.005 + 1]$$

$$= (80000 - 6M)(1.005)^2 - M(2.005)$$

$$(iii) A_{120} = (80000 - 6M)(1.005)^{114} - M[1 + 1.005 + \dots + 1.005^{113}]$$

$$G.P. \quad a=1$$

$$r = 1.005$$

$$n = 114$$

$$= (80000 - 6M)(1.005)^{114} - M \left[\frac{a(r^n - 1)}{r - 1} \right]$$

$$= (80000 - 6M)(1.005)^{114} - M \left[\frac{1.005^{114} - 1}{0.005} \right] \quad (ii)$$

$$= (80000 - 6M)(1.005)^{114} - 200M(1.005^{114} - 1)$$

$$i) A_{120} = 0 \quad \text{Find } M$$

$$0 = 80000(1.005)^{114} - M[6(1.005)^{114} + 200(1.005^{114} - 1)] \quad (iii)$$

$$M = 80000(1.005)^{114} \div 163.746 \dots$$

$$= \$862.681 \dots$$

$\therefore$ repayment is \$863.

(i)

Let X be the intersection of the diagonals.

$\angle XAD = \angle XAC = \theta$ [property of rhombi]

$$AX = 2\cos\theta \Rightarrow AC = 4\cos\theta$$

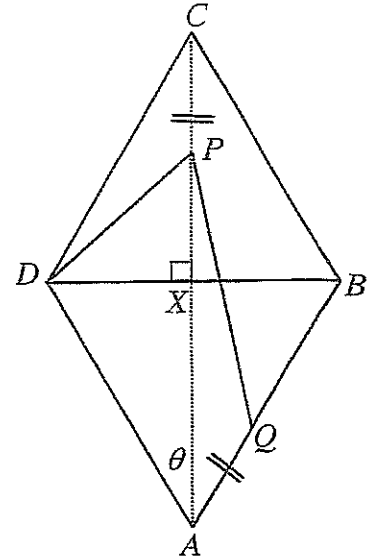
$$\therefore AP = 4\cos\theta - x$$

The shaded area is the sum of triangles ADP and A

$$S = \frac{1}{2} \times 2 \times (4\cos\theta - x)\sin\theta + \frac{1}{2} \times (4\cos\theta - x) \times x$$

$$= \frac{\sin\theta}{2} (4\cos\theta - x)(x + 2)$$

[NB S is a concave down parabola in x]



$$S = \frac{\sin\theta}{2} [8\cos\theta + (4\cos\theta - 2)x - x^2]$$

$$\frac{dS}{dx} = \frac{\sin\theta}{2} [(4\cos\theta - 2) - 2x] = \sin\theta(2\cos\theta - 1 - x)$$

$$\therefore \frac{dS}{dx} = 0 \Rightarrow x = 2\cos\theta - 1 \quad [\because \sin\theta \neq 0]$$

$$\frac{dS}{dx} = \sin\theta(2\cos\theta - 1 - x)$$

$$\therefore \frac{d^2S}{dx^2} = -\sin\theta \quad \left[< 0 \text{ for } 0 < \theta < \frac{\pi}{2} \right]$$

$$(iv) \quad \theta = \frac{\pi}{6}$$

$$\frac{dS}{dx} = 0 \Rightarrow x = 2\cos\left(\frac{\pi}{6}\right) - 1 = \sqrt{3} - 1$$

$$\frac{d^2S}{dx^2} < 0 \Rightarrow S \text{ is a maximum}$$

$$AC = 4\cos\left(\frac{\pi}{6}\right) = 2\sqrt{3}$$

$$\therefore \frac{PC}{AC} = \frac{\sqrt{3} - 1}{2\sqrt{3}}$$